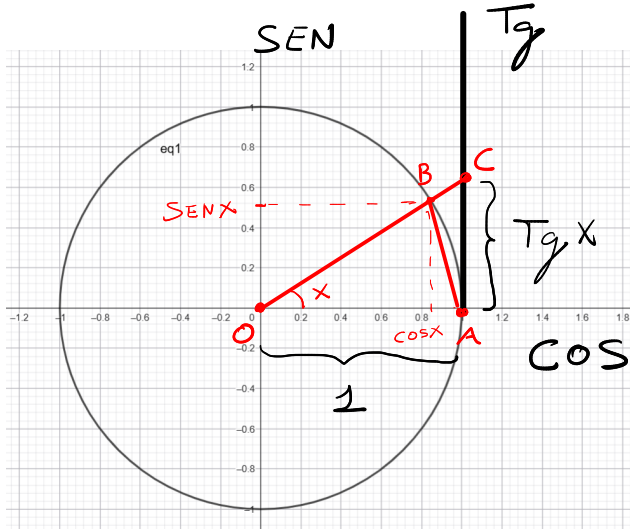


LIMITE TRIGONOMÉTRICO FUNDAMENTAL

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

$$x \rightarrow 0 \Rightarrow x \approx 0 \Rightarrow \frac{\sin 0}{0} = \frac{0}{0} \quad !? \quad (\text{INDETERMINAÇÃO})$$



EM TERMOS DE ÁREA

$$\text{Área } \Delta OAB < \text{Área } \Delta OAB < \text{Área } \Delta OAC$$

$$\frac{1 \cdot \sin x}{2} < \frac{x}{2} < \frac{1 \cdot \tan x}{2}$$

Logo,

$$\frac{\sin x}{2} < \frac{x}{2} < \frac{\tan x}{2}$$

Ou seja: $\sin x < x < \tan x$, mas $\tan x = \frac{\sin x}{\cos x}$.

ENTÃO:

$$\sin x < x < \frac{\sin x}{\cos x}$$

$$\sin x < x$$

$$\frac{\sin x}{x} < 1$$

$$x < \frac{\sin x}{\cos x}$$

$$x \cdot \cos x < \sin x$$

$$\cos x < \frac{\sin x}{x}$$

$$\cos x < \frac{\sin x}{x} < 1 \Rightarrow \lim_{x \rightarrow 0} \cos x = 1$$

$$\Rightarrow 0,999 \leq f(x) < 1$$

$$L = 1$$

Exemplo 1. Use a Definição (2.6) para provar que

$$\lim_{x \rightarrow 4} \underbrace{\frac{1}{2}(3x-1)}_{f(x)} = \frac{11}{2} \quad L$$

$$f(x) = \frac{1}{2}(3x-1)$$

$$\lim_{x \rightarrow 4} f(x) = \frac{1}{2}(3 \cdot 4 - 1) = \frac{1}{2} \cdot 11 = \frac{11}{2} \quad \left(\begin{array}{l} \text{MÉTODO} \\ \text{PA SUBSTITUIÇÃO} \\ \text{DIRETA} \end{array} \right)$$

$$x \rightarrow 4 \Rightarrow x \neq 4$$

LIMITE PELA DEF.

Seja $\varepsilon > 0$:

$$|f(x) - L| < \varepsilon$$

$$\left| \frac{1}{2}(3x-1) - \frac{11}{2} \right| < \varepsilon \Leftrightarrow \left| \frac{1}{2} \cdot ((3x-1) - 11) \right| < \varepsilon$$

$$|a \cdot b| = |a| \cdot |b|$$

$$\Leftrightarrow \left| \frac{1}{2} \right| \cdot \left| (3x-1) - 11 \right| < \varepsilon \Leftrightarrow \frac{1}{2} \cdot |3x-12| < \varepsilon$$

$$\Leftrightarrow \frac{1}{2} \cdot |3| \cdot |x-4| < \varepsilon \Leftrightarrow \frac{3}{2} \cdot |x-4| < \varepsilon$$

$$\Leftrightarrow |x-4| < \underbrace{\frac{\varepsilon \cdot 2}{3}}_s$$

Dado $\varepsilon > 0$, faça $s = \frac{2}{3}\varepsilon$.

Nesse caso

$$x \in D_f \wedge |x-4| < s \Rightarrow |f(x) - \frac{11}{2}| < \varepsilon$$

$$\text{Portanto, } \lim_{x \rightarrow 4} f(x) = \frac{11}{2}$$